

HAFIZA LAIBA FAISAL

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PROFILE

Electrical Engineering undergraduate with experience in power systems, embedded systems, aerospace engineering, and artificial intelligence. Passionate about large-scale infrastructure, energy technologies, and intelligent engineering solutions — with hands-on experience in transmission planning, IoT energy systems, sensor fusion, and multidisciplinary research projects. Recipient of multiple national awards including 1st Place in Pakistan's National CanSat Competition and Student of the Year.

EDUCATION

Bachelor of Electrical Engineering

Expected 2027

Lahore College for Women University (LCWU), Pakistan

Relevant Coursework: Signals & Systems · Control Systems · Power Electronics · Embedded Systems · Digital Logic Design · Probability Methods in Engineering

INDUSTRY EXPERIENCE

Power System Planning Intern | National Transmission & Despatch Company (NTDC) Jun 2025 – Aug 2025

Competitive Selection — Top 12% Nationwide (Grid Girls Pakistan, ADB-funded)

- Conducted power flow studies and N-1 contingency analysis for Pakistan's national high-voltage transmission network using PSS®E.
- Performed dynamic stability analysis supporting long-range transmission planning and grid reliability assessments.

AI Full Stack Software Engineer | PenTutor — Production Educational AI Platform Jan 2026 – Present

- Architected LLM-powered student support systems and privacy-aware AI content moderation workflows serving real users.
- Designed role-based access control, scalable Django backend services, and deployment automation pipelines.
- Contributed to platform reliability and API development across production educational infrastructure.

HONOURS & AWARDS

- **1st Position — Pakistan's First National CanSat Competition (SUPARCO), 2025**
- Student of the Year — Lahore College for Women University
- Grid Girls Pakistan — Selected among top 12% nationwide (ADB-funded national initiative)
- Innovista AI Hackathon — Winner, FloodGuard AI
- Pakathon — 3rd Position (National Entrepreneurship Competition)
- International Recognition — NeuroShield, PeerBridge Mental Health Hacks

SELECTED PROJECTS

SUPARCO National CanSat Mission System 🏆 *1st Place — National Competition*

- Developed and experimentally validated real-time 2D/3D trajectory reconstruction deployed on physical CanSat hardware during live national competition missions.
- Implemented full sensor fusion pipeline: accelerometer, gyroscope, magnetometer, GPS, and telemetry — with orientation estimation, geolocation, and impact prediction modules.

Smart Energy Audit Monitoring System *Embedded Systems & IoT*

- Developed an IoT-based smart energy monitoring and prepaid management system using ESP32 and PZEM-004T with real-time power measurement, anomaly detection, remote load control, and energy optimization capabilities.

FloodGuard AI — Disaster Management Platform *Innovista Hackathon Winner*

- Developed a multi-agent flood forecasting and evacuation planning system integrating satellite data, weather feeds, and GIS-based risk modelling.

CircuitSathi — AI Circuit Analysis & Debugging Assistant *Educational Technology*

- Developed an AI-powered engineering assistant for circuit analysis, fault detection, debugging, and electrical engineering education.

NeuroShield — EEG-Powered Addiction Recovery Platform *International Recognition, PeerBridge*

- Developed an AI-assisted addiction recovery platform integrating real-time EEG signal analysis with intelligent behavioural support systems.

HealthLink360 — Enterprise Healthcare Intelligence Platform *Capstone Project*

- Designed an AI-powered healthcare ecosystem integrating emergency response, pharmacy management, patient care coordination, and national health reporting.

RESEARCH EXPERIENCE

EEG-Based Cognitive Load & Confusion Detection | *Lahore College for Women University*

Undergraduate Research

- Developed explainable ML models (XGBoost, SHAP, LIME) for cognitive state classification from EEG signals, targeting adaptive learning applications.
- Investigated physiological markers of human confusion and cognitive overload through signal processing and feature analysis.

TECHNICAL COMPETENCIES

Engineering & Systems: PSS®E · Power Flow Analysis · N-1 Contingency Analysis · Dynamic Stability Analysis · Transmission Planning · Embedded Systems (ESP32, Arduino) · Sensor Fusion · Telemetry Systems · ROS2 · Gazebo · MATLAB · Control Systems

AI & Machine Learning: Explainable AI (SHAP, LIME, XGBoost) · LLM Applications · Multi-Agent Systems · NLP · Computer Vision · EEG Signal Processing · RAG

Software Engineering: Python · C/C++ · JavaScript · Django · FastAPI · React · REST APIs · PostgreSQL · MongoDB · Linux

Domains: Power Systems & Grid Infrastructure · Healthcare Technology · Disaster Management · IoT & Energy Systems · GIS & Geospatial Analysis · Aerospace Engineering · Open Source Development

SELECTED TECHNICAL HIGHLIGHTS

- National aerospace competition winner — real hardware deployment under competition conditions (SUPARCO)
- Industry experience in transmission system planning at NTDC (national grid operator)
- 3+ years of software and AI development experience across research, production systems, and competitions
- Production engineering experience at PenTutor (LLM-powered educational AI platform)
- Open-source contributor: Blackwell OS (browser/desktop components) and MIPSDev (educational tooling)
- Led multidisciplinary engineering teams across aerospace, AI, embedded systems, and healthcare domains

CERTIFICATIONS

- Huawei HCIA-AI — Huawei Certification Program
- Machine Learning Professional Certificate — IBM
- Multi AI Agent Systems with AutoGen — DeepLearning.AI
- AI Product Management — Microsoft
- Azure AI Apps & Agents — Microsoft Azure

LEADERSHIP & COMMUNITY ENGAGEMENT

Engineering Team Lead | *Multiple Projects*

- Led multidisciplinary teams across aerospace, AI, embedded systems, and healthcare domains from design through live competition deployment.
- Coordinated requirements scoping, technical implementation, testing, and formal presentations to faculty and competition judges.

AI Workshop Facilitator & Student Mentor | *University Community*

- Designed and facilitated Generative AI workshops for university students; mentored participants on applied AI, research pathways, and project development.
- Provided individual guidance on technical projects and career planning in AI and engineering research.

Open Source Contributions

- Blackwell OS — Browser components and desktop ecosystem features; code review and collaborative release workflows.
- MIPSDev — Validation testing and tooling improvements for VS Code-based MIPS educational development extension.

RESEARCH INTERESTS

Smart power systems, intelligent embedded systems, renewable energy integration, energy system optimization, AI-assisted infrastructure monitoring, autonomous aerospace systems, and human-centred artificial intelligence.

References available upon request